

```
"""
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```
Safe Haven Systems Design Document Generator (Python)  
Project Safe Haven | CIS-212 | Belhaven University  
By: Gramel Rosell | April 2026  
"""
```

```
from docx import Document  
from docx.shared import Pt, Inches  
from docx.enum.text import WD_ALIGN_PARAGRAPH  
from docx.oxml.ns import qn  
from docx.oxml import OxmlElement  
import os
```

```
doc = Document()
```

```
# Default font  
style = doc.styles["Normal"]  
style.font.name = "Arial"  
style.font.size = Pt(12)
```

```
# Page setup  
for section in doc.sections:  
    section.page_width = Inches(8.5)  
    section.page_height = Inches(11)  
    section.top_margin = Inches(1)  
    section.bottom_margin = Inches(1)  
    section.left_margin = Inches(1)  
    section.right_margin = Inches(1)
```

```
def add_para(text, bold=False, italic=False, size=12, align=None, after=Pt(8)):  
    para = doc.add_paragraph()  
    para.paragraph_format.space_after = after  
    para.paragraph_format.space_before = Pt(0)  
    para.paragraph_format.line_spacing = Pt(18)  
    if align:  
        para.alignment = align  
    run = para.add_run(text)  
    run.font.name = "Arial"  
    run.font.size = Pt(size)  
    run.font.bold = bold  
    run.font.italic = italic  
    return para
```

```
def bold_start(bold_text, normal_text):  
    para = doc.add_paragraph()  
    para.paragraph_format.space_after = Pt(8)  
    para.paragraph_format.space_before = Pt(0)  
    para.paragraph_format.line_spacing = Pt(18)  
    r1 = para.add_run(bold_text)  
    r1.font.name = "Arial"  
    r1.font.size = Pt(12)  
    r1.font.bold = True  
    r2 = para.add_run(normal_text)  
    r2.font.name = "Arial"  
    r2.font.size = Pt(12)  
    return para
```

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def blank(n=1):
    for _ in range(n):
        p = doc.add_paragraph()
        p.paragraph_format.space_after = Pt(4)
        p.paragraph_format.space_before = Pt(0)

def setup_header_footer(sec):
    hdr = sec.header
    hdr.is_linked_to_previous = False
    hp = hdr.paragraphs[0]
    hp.text = ""
    r = hp.add_run("Project Safe Haven \u2014 Systems Design Document")
    r.font.name = "Arial"
    r.font.size = Pt(9)
    r.font.italic = True

    ftr = sec.footer
    ftr.is_linked_to_previous = False
    fp = ftr.paragraphs[0]
    fp.alignment = WD_ALIGN_PARAGRAPH.CENTER
    fp.text = ""
    r = fp.add_run("Page ")
    r.font.name = "Arial"
    r.font.size = Pt(9)
    fc1 = XmlElement("w:fldChar")
    fc1.set(qn("w:fldCharType"), "begin")
    r2 = fp.add_run()
    r2._r.append(fc1)
    it = XmlElement("w:instrText")
    it.set(qn("xml:space"), "preserve")
    it.text = " PAGE "
    r3 = fp.add_run()
    r3.font.name = "Arial"
    r3.font.size = Pt(9)
    r3._r.append(it)
    fc2 = XmlElement("w:fldChar")
    fc2.set(qn("w:fldCharType"), "end")
    r4 = fp.add_run()
    r4._r.append(fc2)

# =====
# TITLE PAGE
# =====

blank(5)
add_para("Project Safe Haven", bold=True, size=26, align=WD_ALIGN_PARAGRAPH.CENTER,
after=Pt(12))
add_para("Systems Design Document", size=18, align=WD_ALIGN_PARAGRAPH.CENTER,
after=Pt(24))
blank(1)
add_para("CIS-212", size=14, align=WD_ALIGN_PARAGRAPH.CENTER, after=Pt(8))
add_para("Belhaven University", size=14, align=WD_ALIGN_PARAGRAPH.CENTER,
after=Pt(16))
blank(1)
add_para("By: Gramel Rosell", size=14, align=WD_ALIGN_PARAGRAPH.CENTER,

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after=Pt(8))
add_para("February 2026", size=14, align=WD_ALIGN_PARAGRAPH.CENTER, after=Pt(16))
blank(1)
add_para("Institutional Grant Request: $2,000,000", bold=True, size=14,
align=WD_ALIGN_PARAGRAPH.CENTER)

# =====
# CONTENT – CONTINUOUS FLOW, NO BREAKS
# =====

doc.add_page_break()
setup_header_footer(doc.sections[-1])

# Section 1
bold_start(
    "1. User Interface for Suicide Prevention. ",
    "The Safe Haven application is designed as a comprehensive, student-centered "
    "digital wellness platform that provides Belhaven University students with "
    "immediate access to mental health resources, self-monitoring tools, and crisis
"
    "intervention services. The user interface follows a mobile-first responsive "
    "design philosophy, ensuring that every student can access critical support
from "
    "any device at any time. The interface uses calming color palettes, clear "
    "navigation, and accessible design to reduce cognitive load during moments of "
    "emotional distress."
)

add_para(
    "The application is organized around four primary user actions. First, students
"
    "can input their mental health issues by logging their current emotional state,
"
    "stressors, and concerns through structured intake forms and free-text entry "
    "fields. The system uses validated clinical instruments including the PHQ-9 for
"
    "depression screening and the GAD-7 for anxiety assessment. All submissions are
"
    "encrypted and stored under the zero-identifier privacy model. Second, students
"
    "can track their counseling by viewing scheduled appointments, session history,
"
    "treatment plan progress, and secure messaging with assigned counselors. The "
    "tracker supports both in-person and telehealth sessions with automated
reminders "
    "sent before each appointment. Third, students can view their progress through
a "
    "personalized dashboard that displays mood trend visualizations, journaling "
    "frequency, goal completion rates, and clinical assessment score trajectories "
    "over daily, weekly, and monthly timeframes. Fourth, students can set personal
"
    "goals across categories such as emotional regulation, academic performance, "
    "social engagement, and physical activity, with sub-milestones, deadline "
    "tracking, and automated progress notifications."
)

add_para(
    "The platform organizes its counseling infrastructure into five service

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modules. "
    "Mental Health Counseling provides individual one-on-one sessions with licensed
    "
    "counselors specializing in trauma-informed care, cognitive behavioral therapy,
    "
    "and culturally responsive treatment, delivered through both synchronous video
    "
    "and asynchronous secure messaging. Group Counseling offers facilitated group "
    "therapy sessions of six to twelve participants organized by theme, including "
    "anxiety management, grief and loss, identity exploration, substance misuse,
and "
    "academic stress. Therapy provides specialized evidence-based modalities "
    "including Dialectical Behavior Therapy, Eye Movement Desensitization and "
    "Reprocessing, and Acceptance and Commitment Therapy with structured treatment
    "
    "plans and progress tracking integrated into the analytics engine. The 24-Hour
    "
    "Crisis Hotline delivers round-the-clock crisis response accessible with a "
    "single tap from any screen, with automated safety planning, real-time risk "
    "assessment, and escalation pathways to campus safety, emergency services, or "
    "the 988 Suicide and Crisis Lifeline. Progress and Analytics provides
dashboards "
    "showing PHQ-9 and GAD-7 score trends, mood patterns, session attendance, and "
    "goal completion rates, with an insight engine that generates analytical "
    "improvement recommendations based on longitudinal data."
)

add_para(
    "The goal-setting module allows students to establish forward-looking goals "
    "using SMART methodology across wellness dimensions including emotional "
    "resilience, academic achievement, and career development. A career research "
    "integration encourages students to explore their professional fields as a "
    "protective factor against burnout. The notification engine monitors milestones
    "
    "in real-time and sends personalized alerts when goals are completed, deadlines
    "
    "approach, or engagement patterns drop, delivered through push notifications, "
    "in-app alerts, and optional email with positive reinforcement messaging."
)

# Section 2
bold_start(
    "2. Essential Features. ",
    "The Minimum Viable Product for Safe Haven includes eight core feature modules
    "
    "validated through stakeholder interviews, user story mapping, and prioritized
    "
    "backlog creation. User Registration and Authentication enables secure
onboarding "
    "using QR code scanning for campus-based login and logout, with single sign-on
    "
    "integration to Belhaven\u2019s student information system, multi-factor "
    "authentication for sensitive data access, and automatic session timeout after
    "
    "periods of inactivity. The QR code system enables frictionless check-in at "
    "physical counseling locations while maintaining digital session continuity."
)

add_para(

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"Mood Tracking provides a daily logging interface where students record their " "emotional state using visual mood selectors, numerical rating scales aligned " "with clinical instruments, and optional free-text context notes. Mood data feeds " "directly into the predictive analytics engine for risk classification, and " "students can view their history through interactive charts. Daily Journaling " "offers a private, encrypted space where students write reflective entries with " "guided prompts based on therapeutic modalities such as gratitude journaling, " "cognitive restructuring, and behavioral activation. Journal entries are stored " "under the zero-identifier privacy model and are never accessible to counselors " "or administrators without explicit student consent."

add_para("Accessibility and Integration with Resources ensures full WCAG 2.1 AA " "compliance including screen reader support, keyboard navigation, high-contrast " "mode, and adjustable font sizes. The resource integration module connects " "students with campus mental health services, external provider directories, " "crisis resources, and self-help content through a unified navigation interface." "Privacy and Security implements a zero-identifier architecture that separates " "student identity data from clinical and behavioral data at the database level, " "with AES-256 encryption at rest, TLS 1.3 encryption in transit, role-based " "access control, comprehensive audit logging, and full FERPA and HIPAA compliance.")

add_para("User Feedback and Contact Options provides an in-app feedback system for " "ratings, anonymous submissions, post-session satisfaction surveys, and " "categorized contact routing to appropriate departments. Push Notifications " "delivers configurable reminders for counseling appointments, mood logging " "prompts, goal milestone alerts, wellness tips, and campus mental health events, " "with students having granular control over notification frequency and timing " "and crisis-related alerts prioritized above all other types. Anonymous Peer " "Support Chat enables moderated conversations where students share experiences " "without stigma, with automated content safety filtering and trained peer " "moderator tools.")

Section 3

bold_start("3. Steps to Build and Validate the MVP. ", "The development process follows four phases. The Breakdown Analysis phase " "established that Safe Haven users will be able to log in and out by scanning a " "QR code, request email communications, track their mood, write daily journal " "entries, access resources with full accessibility support, maintain privacy and " "security, submit feedback and contact options, and receive push notifications."

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    "During this phase, the team completed stakeholder interviews, user story "
    "mapping, and prioritized backlog creation. The done rule required that all
user "
    "stories be clearly defined, validated by stakeholders, and prioritized for "
    "development."
)

add_para(
    "The Design and Development phase translated user stories into application "
    "designs and code meeting design acceptance criteria. Wireframes and prototypes
"
    "were reviewed and approved by stakeholders before development of MVP features
"
    "commenced. The prototyping process used three stages: low-fidelity wireframes
"
    "created in Figma for information architecture, mid-fidelity interactive "
    "prototypes for usability testing, and high-fidelity mockups with the final "
    "visual design system. The design system uses calming blues and greens,
generous "
    "whitespace, a minimum 16-pixel base font size, and 44-by-44-pixel touch
targets "
    "for accessibility compliance."
)

# Section 4
bold_start(
    "4. Technology Stack. ",
    "The frontend is built using React Native, enabling cross-platform deployment "
    "to both iOS and Android from a single codebase with native-quality
performance. "
    "TypeScript provides strict typing across the entire frontend for reduced
runtime "
    "errors and improved maintainability. The react-native-chart-kit library
renders "
    "mood trend graphs, clinical score trajectories, goal progress charts, and "
    "counseling analytics within the student dashboard. React Navigation provides "
    "stack, tab, and drawer navigation patterns with deep linking support for push
"
    "notification routing."
)

add_para(
    "The backend infrastructure uses Firebase for real-time database
synchronization "
    "and user authentication, handling QR code verification, email-based account "
    "recovery, and session management. Node.js with Express serves as the primary "
    "API server for RESTful services handling user data management, counseling "
    "scheduling, goal tracking, and resource catalog operations with rate limiting,
"
    "input validation, and request logging. Python Flask powers a dedicated "
    "predictive analytics microservice that processes PHQ-9 and GAD-7 assessment "
    "scores, academic performance indicators, and lifestyle inputs through machine
"
    "learning classification models that assign students to Healthy, At-Risk, or "
    "Struggling risk tiers. PostgreSQL serves as the primary relational database "
    "with row-level encryption implementing the zero-identifier privacy model "
    "through a separation architecture where identity tables and clinical data "
    "tables are linked only through encrypted reference tokens. Redis provides "
    "in-memory caching for session management and rate limiting to ensure "

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    "sub-200-millisecond API response times. The cloud infrastructure runs on AWS "
    "GovCloud or Azure for HIPAA-compliant hosting, with GitHub Actions for "
    "automated testing and deployment, and Prometheus with Grafana for system "
    "health monitoring."
)

add_para(
    "The system implements a three-tier distributed architecture. The Client Layer "
    "consists of the React Native mobile application, a progressive web "
    "application, "
    "and the QR scanner module, responsible for user interface rendering, local "
    "data "
    "caching, authentication, and push notification handling. The Application Layer "
    "includes the Node.js API gateway, Flask analytics service, and Firebase cloud "
    "functions, handling business logic, API routing, clinical assessment "
    "processing, "
    "risk classification, notification dispatch, and session management. The Data "
    "Layer encompasses PostgreSQL with encryption, Redis cache, and Firebase "
    "Realtime "
    "Database, providing persistent clinical data storage, session caching, and "
    "real-time synchronization."
)

# Section 5
bold_start(
    "5. Testing and Deployment. ",
    "The testing strategy follows a comprehensive multi-layered approach. Unit "
    "testing uses Jest for JavaScript and TypeScript modules and pytest for Python "
    "analytics services, targeting 80 percent coverage with 100 percent required "
    "for security-critical functions including authentication, data encryption, and "
    "clinical data processing. Integration testing verifies end-to-end data flows, "
    "confirming that entries submitted from the mobile app correctly propagate "
    "through the API layer, are encrypted in the database, trigger analytics "
    "processing, and display accurately in dashboard visualizations. Backend "
    "testing "
    "uses the Firebase Emulator Suite for local testing of authentication flows, "
    "real-time database operations, and cloud function triggers without impacting "
    "production data. User Acceptance Testing involves 50 to 100 Belhaven student "
    "volunteers testing the application for two weeks to validate clinical "
    "instruments, accessibility features, and real-world usability through surveys "
    "and focus groups. Security testing includes third-party OWASP Top 10 "
    "vulnerability assessment, penetration testing, authentication bypass testing, "
    "and HIPAA and FERPA compliance verification, requiring zero critical findings "
    "before deployment. The done rule requires that all tests pass with no critical "
    "issues and that application usability meets user expectations."
)

add_para(
    "The deployment follows a staged process. The application first deploys to a "

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"staging environment mirroring production for load testing with 3,500 simulated
"
"concurrent connections and full smoke testing. A beta release then goes to 200
"
"selected students across all demographics for a 30-day monitored period. "
"Production deployment releases the application to the Google Play Store and "
"Apple App Store using a blue-green strategy enabling instant rollback, with "
"continuous monitoring for the first 72 hours and daily monitoring for the
first "
"month. Post-launch monitoring uses Prometheus and Grafana dashboards tracking
"
"response times, error rates, engagement metrics, and clinical data integrity,
"
"with a 24/7 on-call rotation for the initial 90 days."
)

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add_para(
"The RESTful API provides endpoints including POST to /api/auth/qr-login for "
"QR code authentication, POST to /api/mood/log for submitting mood entries, GET
"
"to /api/mood/history for retrieving mood trends, POST to /api/journal/entry
for "
"creating journal entries, POST to /api/assessment/phq9 for submitting clinical
"
"assessments, GET to /api/analytics/risk-score for counselor access to risk "
"classifications, POST to /api/crisis/initiate for triggering crisis protocols,
"
"and GET to /api/dashboard/admin for institutional analytics."
)

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Section 6

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bold_start(
"6. Scale Up. ",
"Once the MVP is launched and stabilized, the development roadmap expands to "
"include advanced capabilities. Advanced Mood Analysis will integrate natural "
"language processing models to analyze journal entries for sentiment patterns,
"
"identifying early warning indicators beyond standard clinical scales and
routing "
"alerts to counselors. Community Features will expand the peer support system "
"into moderated spaces organized by topic with trained moderators and automated
"
"content safety filtering. Wearable Device Integration will connect to Apple "
"HealthKit, Google Fit, and Fitbit to incorporate sleep quality, heart rate "
"variability, and physical activity data into the predictive model. The "
"Institutional Data Environment will provide administrators with anonymized "
"dashboards showing campus-wide mental health trends, intervention
effectiveness, "
"and outputs for grant reporting and IPEDS benchmarking."
)

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Conclusion

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bold_start(
"Conclusion. ",
"The Safe Haven Initiative represents a transformative two-million-dollar "
"investment in Belhaven University\u2019s mental health infrastructure. This "
"systems design document has outlined the complete technical architecture, user
"
"interface design, feature specifications, technology stack, testing strategy,

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"
    "and deployment plan required to deliver a production-ready digital wellness "
    "platform within the 24-month implementation window."
)

add_para(
    "The development team proposes using Kanban methodology to manage the project,
    "
    "ensuring continuous delivery through iterative development while maintaining "
    "flexibility to respond to stakeholder feedback. The platform delivers four
core "
    "commitments: comprehensive resource access through a clear, organized catalog
    "
    "integrating campus and external mental health services into a single
interface; "
    "anonymous support options including anonymous chat, de-identified mood
tracking, "
    "and zero-identifier journals that reduce stigma and encourage engagement; "
    "self-help tools providing guided journaling, mood tracking with personalized "
    "insights, and goal-setting frameworks for consistent support between
counseling "
    "sessions; and a centralized platform that unifies counseling scheduling,
crisis "
    "intervention, peer support, clinical assessments, progress analytics, and "
    "resource discovery, eliminating fragmentation across disconnected systems."
)

add_para(
    "At Safe Haven, all staff members are our first responders. Counseling staff
are "
    "available 24 hours a day, and the digital platform extends that commitment to
    "
    "every moment a student reaches for support. Expected outcomes within 24 months
    "
    "include a 50 percent reduction in counseling wait times, a 25 percent increase
    "
    "in help-seeking behavior among underserved student groups, 10 to 15 percent "
    "improvement in academic performance among high-risk students, measurable "
    "reduction in crisis incidents, and $200,000 to $400,000 in annual retention "
    "revenue projected to fully offset the investment within three years."
)

blank(1)
add_para("Prepared by: Gramel Rosell, Belhaven University, February 2026",
        italic=True, align=WD_ALIGN_PARAGRAPH.CENTER, after=Pt(4))
add_para("Data Sources: SAMHSA MH-CLD, CDC BRFSS, IPEDS, NCES DataLab",
        italic=True, align=WD_ALIGN_PARAGRAPH.CENTER, after=Pt(4))

# =====
# SAVE
# =====

output =
"/sessions/upbeat-wonderful-cori/mnt/outputs/Safe_Haven_Systems_Design_Document.doc
x"
doc.save(output)
size = os.path.getsize(output) / 1024
print(f"Document created: {output}")

```

```
print(f"Size: {size:.1f} KB")  
print("Python script completed successfully.")
```